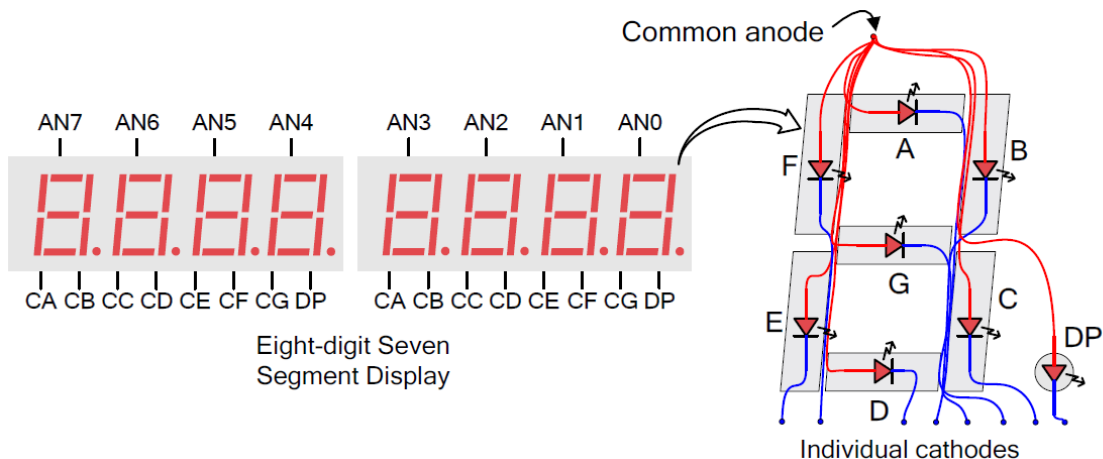


Lab2. 7-segment를 사용한 십진수 표시

다음과 같은 7-segment를 사용하여 lab1에서의 덧셈 결과를 십진수로 나타내기로 한다.



예를 들어 1을 나타낼 때는 b, c 만 '0'이고 나머지 LED는 '1'이다. 나머지 숫자는 다음 표를 참고하기 바람.

	a	b	c	d	e	f	g	dp	An7	An6	~	An0
0	0	0	0	0	0	0	1	1	0	1	1	1
1	1	0	0	1	1	1	1	1	0	1	1	1
2	0	0	1	0	0	1	0	1	0	1	1	1
3	0	0	0	0	1	1	0	1	0	1	1	1
4	1	0	0	1	1	0	0	1	0	1	1	1
5	0	1	0	0	1	0	0	1	0	1	1	1
6	0	1	0	0	0	0	0	1	0	1	1	1
7	0	0	0	1	1	0	1	1	0	1	1	1
8	0	0	0	0	0	0	0	1	0	1	1	1
9	0	0	0	1	1	0	0	1	0	1	1	1

Lab1을 참조하여 프로젝트 이름을 lab2로 하여 새로운 프로젝트를 생성하고 다음 코드를 입력한다.

seg.v 소스파일

```
module seg(a,b,s
    );
input [3:0] a,b;
output [15:0] s;

reg [4:0] sum;
reg [15:0] s;

always @ (a,b)
begin
    sum = a + b;
    s[8:0] = 9'b101111111;
    case (sum)
        0: s[15:9] = 7'b00000001;
        1: s[15:9] = 7'b10011111;
        2: s[15:9] = 7'b0010010;
        3: s[15:9] = 7'b0000110;
        4: s[15:9] = 7'b1001100;
        5: s[15:9] = 7'b0100100;
        6: s[15:9] = 7'b0100000;
        7: s[15:9] = 7'b0001101;
        8: s[15:9] = 7'b0000000;
        9: s[15:9] = 7'b0001100;
        default: s[15:9] = 7'b1111111;
    endcase
end
endmodule
```

테스트벤치 파일 **tb.v**

```
module tb;

    // Inputs
    reg [3:0] a;
    reg [3:0] b;

    // Outputs
    wire [15:0] s;

    // for loop
    reg [4:0] i, j;

    // Instantiate the Unit Under Test (UUT)
    seg uut (
        .a(a),
        .b(b),
        .s(s)
    );

    initial begin
        // Initialize Inputs
        a = 0;
        b = 0;
        for (i=0; i<= 15; i=i+1) begin
            for (j=0; j<=15; j=j+1) begin
                #20 $display("%d + %d ==> %b", a, b, s);
                b = b + 1;
            end
            a = a + 1;
        end
    end

endmodule
```

XDC 파일 seg.xdc

```
#virtual clock
```

```
#create_clock -name VCLK -period 10.0 -waveform {0 5.0}
```

```
#set_input_delay 0.0 -clock [get_clocks VCLK] [get_ports {a b}]
```

```
#set_output_delay 0.0 -clock [get_clocks VCLK] [get_ports s]
```

```
set_property PACKAGE_PIN J15 [get_ports {a[0]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {a[0]}]
```

```
set_property PACKAGE_PIN L16 [get_ports {a[1]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {a[1]}]
```

```
set_property PACKAGE_PIN M13 [get_ports {a[2]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {a[2]}]
```

```
set_property PACKAGE_PIN R15 [get_ports {a[3]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {a[3]}]
```

```
set_property PACKAGE_PIN R17 [get_ports {b[0]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {b[0]}]
```

```
set_property PACKAGE_PIN T18 [get_ports {b[1]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {b[1]}]
```

```
set_property PACKAGE_PIN U18 [get_ports {b[2]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {b[2]}]
```

```
set_property PACKAGE_PIN R13 [get_ports {b[3]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {b[3]}]
```

```
#a
```

```
set_property PACKAGE_PIN T10 [get_ports {s[15]}]
```

```
set_property IOSTANDARD LVCMOS33 [get_ports {s[15]}]
```

```
#b
```

```
set_property PACKAGE_PIN R10 [get_ports {s[14]}]
```

```

set_property IOSTANDARD LVCMOS33 [get_ports {s[14]}]
#c
set_property PACKAGE_PIN K16 [get_ports {s[13]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[13]}]
#d
set_property PACKAGE_PIN K13 [get_ports {s[12]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[12]}]
#e
set_property PACKAGE_PIN P15 [get_ports {s[11]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[11]}]
#f
set_property PACKAGE_PIN T11 [get_ports {s[10]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[10]}]
#g
set_property PACKAGE_PIN L18 [get_ports {s[9]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[9]}]
#dp
set_property PACKAGE_PIN H15 [get_ports {s[8]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[8]}]
#an7
set_property PACKAGE_PIN U13 [get_ports {s[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[7]}]
#an6
set_property PACKAGE_PIN K2 [get_ports {s[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[6]}]
#an5
set_property PACKAGE_PIN T14 [get_ports {s[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[5]}]
#an4
set_property PACKAGE_PIN P14 [get_ports {s[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[4]}]
#an3
set_property PACKAGE_PIN J14 [get_ports {s[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[3]}]
#an2
set_property PACKAGE_PIN T9 [get_ports {s[2]}]

```

```
set_property IOSTANDARD LVCMOS33 [get_ports {s[2]}]
#an1
set_property PACKAGE_PIN J18 [get_ports {s[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[1]}]
#an0
set_property PACKAGE_PIN J17 [get_ports {s[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {s[0]}]
```

[도전문제 1] case 문을 if 문으로 대체하라.

[도전문제 2] 7-segment에 16진수 a,b,c,d,e,f를 추가로 나타낼 수 있도록 코드를 수정하라.